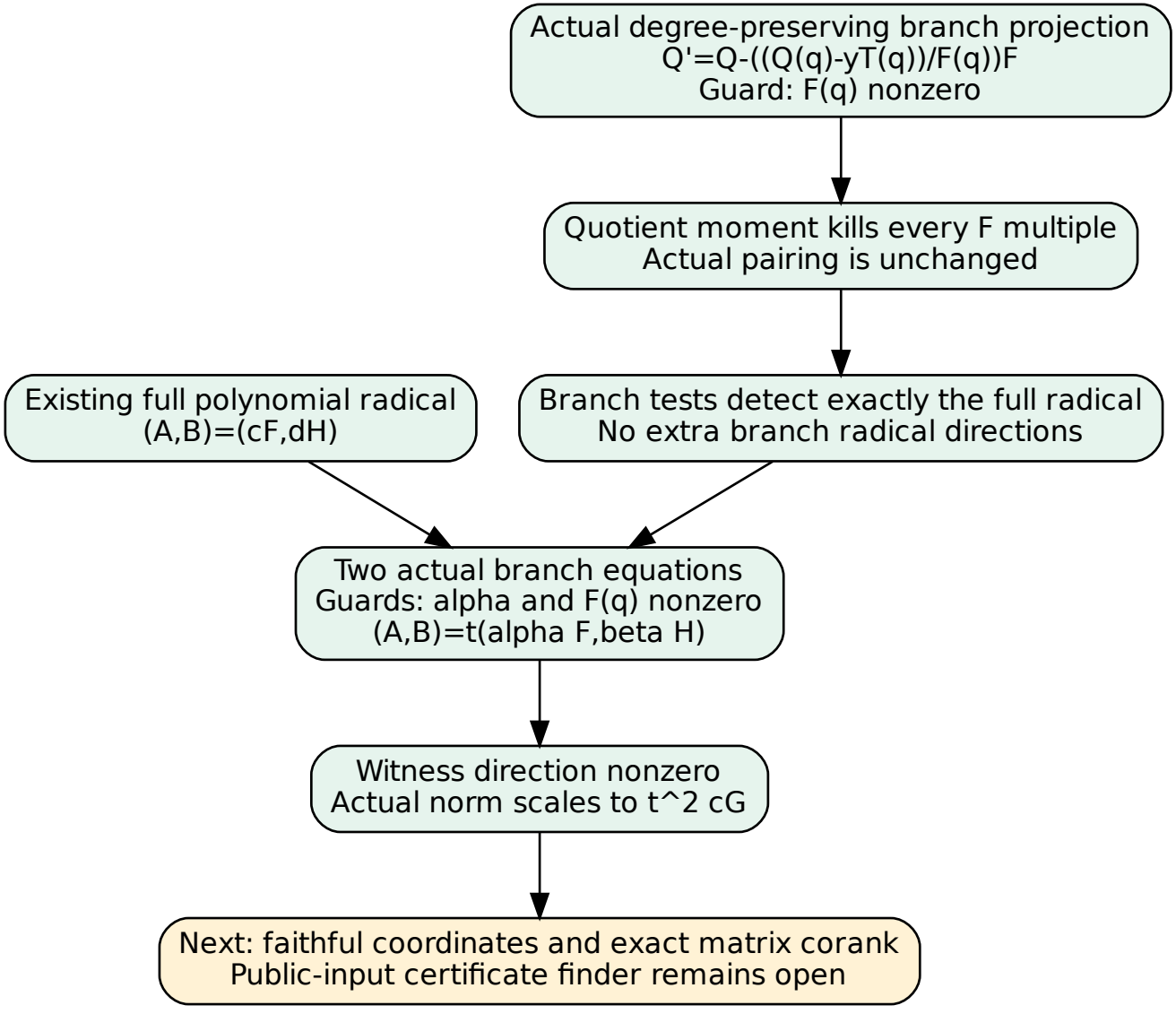
Actual degree-preserving branch projection
Q'=Q-((Q(q)-yT(q))/F(q))F
Guard: F(q) nonzero
Quotient moment kills every F multiple
Actual pairing is unchanged
Existing full polynomial radical
(A,B)=(cF,dH)
Branch tests detect exactly the full radical
No extra branch radical directions
Two actual branch equations
Guards: alpha and F(q) nonzero
(A,B)=t(alpha F,beta H)
Witness direction nonzero
Actual norm scales to t^2 cG
Next: faithful coordinates and exact matrix corank
Public-input certificate finder remains open